



Document Management System



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Department of Child Health

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Title: Management of Biliary Atresia

Introduction:

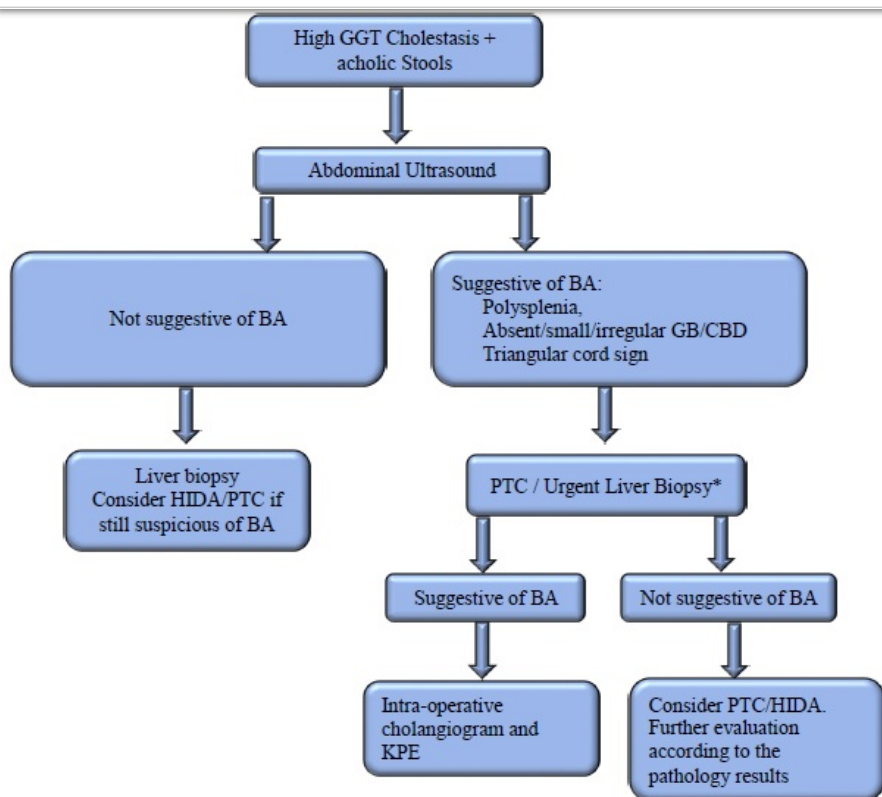
Biliary Atresia: a progressive fibro-obliterative disease of the extrahepatic biliary and to a variable extent the intrahepatic tree that present with jaundice and pale stools in the first 2 months of life. It is the most common cause of liver transplant in children.

Hepato-portoenterostomy or Kasai portoenterostomy (KPE), is a surgery performed on infants with biliary atresia (BA). It is most successful if performed early, preferably before 6 weeks of age.

Cholangiogram is the gold standard diagnostic test for BA. Cholangiogram that shows a normal intra hepatic biliary tree, normal external hepatic ducts and normal drainage into the duodenum excludes the diagnosis.

Biliary Atresia Diagnosis:

- The diagnosis is time-sensitive as current evidence suggest that KPE is most successful when it is done before 60 days of life.
- Any patient with jaundice persists more than 2 weeks should be evaluated for cholestasis with measurement of total and conjugated bilirubin, and that an elevated serum conjugated bilirubin level (conjugated bilirubin levels >17mmol/L) warrants timely consideration for evaluation and referral to a pediatric gastroenterologist.
- All patients with cholestasis should be referred to pediatric gastroenterologist regardless the cause.



*Primary consultant should call Dr Fatma on the day of liver biopsy to ask her to look to the slides as urgent.

Post-operative management during initial hospital stay:

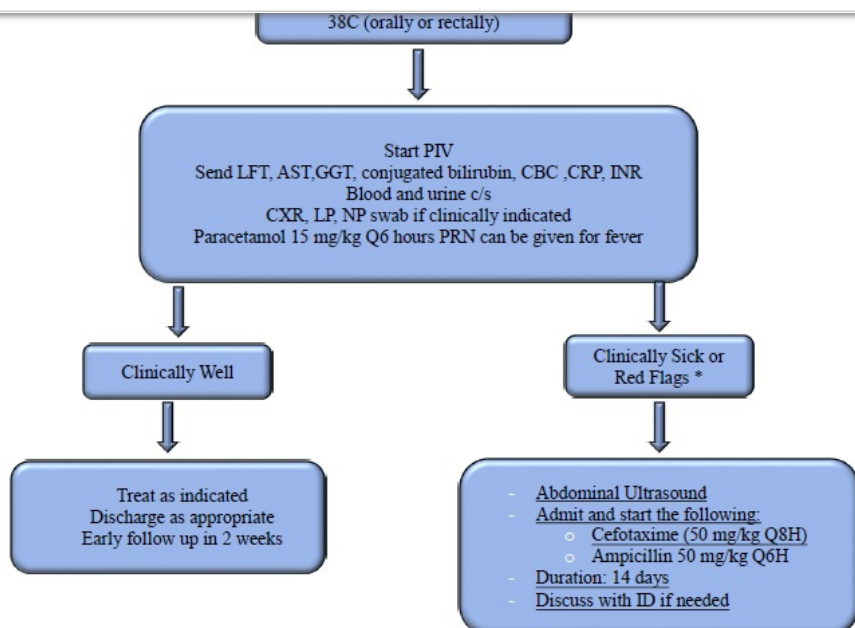
Initial management will be undertaken by the Paediatric Surgical team who will transfer the baby to the Paediatric Gastroenterology team once the baby is stable

- Steroid Therapy:
 - Effectiveness of steroid therapy is controversial and there is no consensus on the regimen when it is started.
 - Steroids to be started on the following groups:
 - High Risk Group:
 - Patients > 60 days of life at KPE
 - Syndromic BA
 - Low Risk Group:
 - Patients who did not meet the above criteria but they have initial response by showing colored stools followed by pale stools later
 - Dose: Prednisolone 5 mg /kg/ day (Prednisolone)
 - Duration: 3-6 weeks. starting from POD 7 to POD 21 then wean off gradually.
 - Steroid therapy effect will be re-evaluated after 2 year based on our patients' outcome and if any new evidence in the literature.
 - Steroid therapy should be decided and written by the gastroenterologist.
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- Antibiotics:
 - Immediate post-operative IV antibiotics to be decided by the surgical team.
 - On completion of IV antibiotics commence long term oral cholangitis prophylaxis with co-trimoxazole (2-5 mg/kg/day as trimethoprim component once daily) for at least 12 months post Kasai.
 - Monitoring:
 - Stool colour should be recorded daily and in particular whether pigment appears in the stool. One stool sample daily should be collected (or pictured) for review by the treating team.
 - Day 1: CBC, LFTs, coagulation screen, electrolytes
 - Thereafter; blood test to be repeated twice a week or as clinically indicated.

- Discharge Medications:
 - Ursodeoxycolic acid 15mg/kg/day divided to BID – secondary hydrophilic bile acid that has choleretic effect by increasing hepatic bile flow and has hepatocyte protective effect.
 - Co-trimoxazole 2-5 mg/kg/day as trimethoprim component once daily- Prophylaxis against cholangitis
 - Vitamin D (cholecalciferol) 1000-2000 iu daily
 - Vitamin E 25 iu/kg daily
 - Vitamin K 2.5 mg daily
- Before discharge make sure that the patient has an appointment with gastroenterology and surgical team as outpatient.
- Educate the parents about the normal stool colored by showing/providing the stool color card.

Management after discharge:

- Clinic appointments:
 - 1st gastroenterology clinic preferred to be after 1-2 weeks from discharge.
 - Patient can be followed monthly thereafter until 6 months of age.
 - Clinic visits should concentrate on the following:
 - Assessing stool color
 - Assessing feeding tolerance and emphasizing adequate growth
 - Examining for signs of portal hypertension
 - Repeat liver injury and function test (ALT, AST, GGT, INR, bilirubin, blood sugar).
 - Fat soluble vitamins should be monitored every 3 months for the first 6 months if the cholestasis is not resolving.
 - Doppler ultrasound after 3 months from the Kasai procedure and earlier if indicated
 - After 6 months if the Kasai procedure is successful (normal bilirubin) then patients can be followed every 3 months in the 1st year then 4-6 monthly.
 - Reasons for more frequent review include, but are not limited to:
 - Poor growth
 - Synthetic liver dysfunction
 - Ascites
 - Portal hypertension and/or gastrointestinal bleeding
 - Recurrent cholangitis
 - Clear need to proceed to liver transplant assessment
- Nutrition:
 - Patients with BA are at high risk of failure thrive secondary to:
 - Malabsorption as result of cholestasis
 - Feeding intolerance due to progressive liver disease and organomegaly
 - Those patients usually require 130-150% of the required caloric needs
 - Nutritionist should be involved in the care from the first visit to the clinic.
 - If the patient is not gaining weight; then the priority is to increase the caloric intake before changing the formula
 - High MCT formula can be considered as 2nd line although the evidence is weak.
 - Early NGT feed should be considered in patients with BA who are not tolerating the required calories orally.
- All immunization should be given for all patients in time without delay.
- Indications for liver transplantation
 - Primary failure of the Kasai procedure- (persistently acholic stools, high bilirubin levels (>6 mg/dL at 3 months))
 - Complication of portal hypertension that are resistant to medical therapy
 - Recurrent cholangitis
 - Intractable pruritis
 - Suspected hepatocellular carcinoma
- Assessment and Management of febrile child with biliary atresia and Kasai portoenterostomy:



*High Risk Red Flags Include (Should be discussed with the gastroenterologist):

- Age < 18 months
- Acholic stools
- Increase GGT and Conjugated bilirubin from baseline
- Decrease albumin from baseline
- Presumed cholangitis in last 6 months

Reference:

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4. Tyraskis A, Davenport M. Steroids after the Kasai procedure for biliary atresia: the effect of age at Kasai portoenterostomy. Pediatr Surg Int. 2016;32:193–200.

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